

Vesicle internalization proceeds via a morphological phase transition

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Abstract

Cells take up large cargo by wrapping it in membrane, yet how a vesicle crosses from partial adhesion to full internalization remains debated. Using simulations we show that internalization is a morphological phase transition of the wrapping membrane. Below a critical adhesion strength the vesicle stays partially wrapped; above it the membrane neck closes abruptly. The transition is controlled by the ratio of adhesion energy to bending rigidity, and line tension at the contact rim sets its sharpness. These findings connect membrane elasticity to the cooperative uptake of nanoparticles and viruses.

Methods

All simulations used the coarse-grained Martini 3 force field with a time step of 20 fs. Membranes were composed of POPC and POPS in a 4:1 ratio, solvated with Martini water and 150 mM NaCl. Temperature was kept at 310 K with the velocity-rescaling thermostat and pressure at 1 bar with a semi-isotropic Parrinello-Rahman barostat. Each vesicle of 20 nm diameter was placed 2 nm above the bilayer. Adhesion strength was tuned by scaling the Lennard-Jones interaction between the vesicle surface beads and the lipid head groups. We ran twelve independent trajectories of 10 microseconds for each adhesion strength using GROMACS 2024. The wrapped fraction was computed from the contact area between vesicle and membrane, averaged over the last 2 microseconds of every trajectory. Bending rigidity was obtained from the undulation spectrum of a tension-free patch. Free energies of wrapping were estimated with umbrella sampling along the vesicle height coordinate.

Results

The wrapped fraction jumps from 0.4 to 1.0 within a narrow window of adhesion strength, a signature of a first-order morphological transition. Near the transition the membrane neck fluctuates between open and closed states, and the free energy profile shows two minima separated by a barrier of about 8 kT. Anionic POPS lipids accumulate at the contact rim and lower the line tension, which shifts the transition to weaker adhesion.

References

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